

Econometrics: Comprehensive Questions and Worked Solutions

Mathematical Foundations, Regression,
Inference, Specification, and Heteroskedasticity

Study companion to Max Heinze's
Econometrics Lecture Notes
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Scope and notation

This study set covers the substantive sections of the supplied 102-page PDF: mathematical and probability preliminaries, introduction to econometrics, simple linear regression, multiple linear regression, testing and inference, additional regression topics, and heteroskedasticity.

Questions include definitions, derivations, calculations, interpretation of the notes' empirical examples, and R exercises. Additional numerical examples are constructed for practice. The solutions correct several errors or overstatements in the source rather than reproducing them.

Throughout:

- N is the sample size.
- K is the number of slope regressors.
- $p = K + 1$ is the total number of coefficients when an intercept is included.
- u_i is a population error, while $\hat{u}_i$ is an estimated residual.
- SSR denotes the residual sum of squares.
- SST denotes the centered total sum of squares.
- SSE denotes the explained sum of squares.

Source references below use the section numbers in the PDF. Examples involving estimated associations should not be interpreted causally without an appropriate identification argument.

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1 Mathematical and Probability Foundations

Source: Sections 0.1–0.2

Question 1: Summation identities

Define the sample mean and prove

$$\sum_{i=1}^N (x_i - \bar{x}) = 0,$$

$$\sum_{i=1}^N (x_i - \bar{x})x_i = \sum_{i=1}^N (x_i - \bar{x})^2,$$

and

$$\sum_{i=1}^N (x_i - \bar{x})^2 = \sum_{i=1}^N x_i^2 - N\bar{x}^2.$$

Worked solution

The sample mean is

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i.$$

Therefore,

$$\sum_i (x_i - \bar{x}) = \sum_i x_i - N\bar{x} = 0.$$

Write $x_i = (x_i - \bar{x}) + \bar{x}$:

$$\sum_i (x_i - \bar{x})x_i = \sum_i (x_i - \bar{x})^2 + \bar{x} \sum_i (x_i - \bar{x}).$$

The last term is zero.

Finally,

$$\begin{aligned} \sum_i (x_i - \bar{x})^2 &= \sum_i x_i^2 - 2\bar{x} \sum_i x_i + N\bar{x}^2 \\ &= \sum_i x_i^2 - N\bar{x}^2. \end{aligned}$$

These identities are central to deriving OLS.

Question 2: Derivatives and the disappearing factor

Differentiate

$$S(b_0, b_1) = \sum_i (y_i - b_0 - b_1 x_i)^2$$

with respect to both parameters. Why can the factor -2 disappear when solving a first-order condition?

Worked solution

By the chain rule,

$$\frac{\partial S}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i),$$

$$\frac{\partial S}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i).$$

When a first-order condition is set equal to zero,

$$-2A = 0 \quad \Longleftrightarrow \quad A = 0,$$

because both sides can be divided by -2 .

Thus, in the no-intercept case corresponding to equations (1.101)–(1.102) in the notes,

$$-2 \sum_i x_i (y_i - b_1 x_i) = 0$$

becomes

$$\sum_i x_i y_i - b_1 \sum_i x_i^2 = 0.$$

The factor is divided out; it is not differentiated away.

Partial differentiation treats other parameters and the observed data as fixed.

Question 3: Logarithms and exponential functions

State the main logarithm rules and their domains. Explain why a small change in $\log x$ approximates a proportional change in x .

Worked solution

For positive x, y ,

$$\begin{aligned} \log(xy) &= \log x + \log y, & \log(x/y) &= \log x - \log y, \\ \log(x^a) &= a \log x, & \log(e^z) &= z. \end{aligned}$$

Here $\log$ means the natural logarithm.

The derivatives are

$$\frac{d \log x}{dx} = \frac{1}{x}, \quad \frac{de^x}{dx} = e^x.$$

Hence

$$d \log x = \frac{dx}{x}.$$

For a finite proportional increase r ,

$$\Delta \log x = \log(1 + r) \approx r$$

when r is small.

For example,

$$\log(1.01) \approx 0.00995 \approx 0.01.$$

The exact inverse conversion is

$$\frac{x_{\text{new}} - x_{\text{old}}}{x_{\text{old}}} = e^{\Delta \log x} - 1.$$

Logarithms of zero or negative real values are not defined in the ordinary real-valued transformation used here.

Question 4: Matrix operations

If X is $N \times p$ and b is $p \times 1$, give the dimensions of Xb , $X'X$, and $X'y$. State the transpose and inverse rules used in regression. When is $X'X$ invertible?

Worked solution

For y of dimension $N \times 1$,

$$Xb : N \times 1, \quad X'X : p \times p, \quad X'y : p \times 1.$$

For conformable matrices,

$$(AB)' = B'A', \quad (A+B)' = A' + B'.$$

For invertible square matrices,

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (A')^{-1} = (A^{-1})'.$$

Matrix multiplication is generally not commutative.

The (i, j) entry of AB is

$$(AB)_{ij} = \sum_{\ell} A_{i\ell} B_{\ell j}.$$

$X'X$ is invertible exactly when X has full column rank. Indeed,

$$c'X'Xc = \|Xc\|^2 > 0$$

for every nonzero c if the columns of X are linearly independent.

Question 5: Random variables, PMFs, PDFs, and CDFs

Distinguish discrete and continuous random variables. Define a PMF, PDF, and CDF. Must a CDF be strictly increasing?

Worked solution

A discrete random variable has finite or countable support. Its probability mass function is

$$p(x) = P(X = x), \quad \sum_x p(x) = 1.$$

For an absolutely continuous random variable, a density $f(x)$ satisfies

$$P(a < X \leq b) = \int_a^b f(x) dx, \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

An individual point has probability zero. A density value is not itself a point probability.

For either type,

$$F(x) = P(X \leq x)$$

is the cumulative distribution function. It is nondecreasing and right-continuous, with limits zero and one at the two infinities.

A CDF need not be strictly increasing: it can be flat on intervals with no probability mass. A discrete CDF has jumps.

Question 6: Expectation, variance, and moments

State the main expectation and variance rules. Find the mean and variance of a Bernoulli variable. Distinguish raw moments, central moments, skewness, and kurtosis.

Worked solution

Linearity of expectation gives

$$\mathbb{E}(aX + bY + c) = a\mathbb{E}X + b\mathbb{E}Y + c.$$

Independence is not required for this identity.

Variance satisfies

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}(X^2) - [\mathbb{E}X]^2, \\ \mathrm{Var}(aX + b) &= a^2 \mathrm{Var}(X), \\ \mathrm{Var}(X + Y) &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2 \mathrm{Cov}(X, Y).\end{aligned}$$

For $D \sim \text{Bernoulli}(\pi)$,

$$\mathbb{E}D = \pi.$$

Since $D^2 = D$,

$$\mathrm{Var}(D) = \pi - \pi^2 = \pi(1 - \pi).$$

The r th raw moment is $\mathbb{E}(X^r)$. The r th central moment is $\mathbb{E}[(X - \mu)^r]$.

Skewness and kurtosis are standardized central moments:

$$\begin{aligned}\text{Skewness} &= \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}, \\ \text{Kurtosis} &= \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}.\end{aligned}$$

Excess kurtosis subtracts 3 from kurtosis.

Question 7: Joint distributions, covariance, and independence

Define a conditional distribution and covariance. Does zero correlation imply independence?

Worked solution

For discrete variables with $P(X = x) > 0$,

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}.$$

For variables with a joint density,

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

where $f_X(x) > 0$.

Independence means that the joint distribution factors into the product of marginal distributions.

Covariance and correlation are

$$\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}X \mathbb{E}Y,$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

The correlation requires positive finite variances.

Independence implies zero covariance when moments exist, but the converse is false.

For example, if X is uniform on $[-1, 1]$ and $Y = X^2$, then

$$\text{Cov}(X, Y) = \mathbb{E}(X^3) - \mathbb{E}X \mathbb{E}(X^2) = 0,$$

yet Y is determined by X .

Zero correlation means no linear association, not no relationship of any kind.

Question 8: Conditional expectations and iterated expectations

State the law of iterated expectations. Show that $\mathbb{E}(u \mid X) = 0$ implies both $\mathbb{E}u = 0$ and $\mathbb{E}[g(X)u] = 0$ for suitable functions g .

Worked solution

The law of iterated expectations states

$$\mathbb{E}[\mathbb{E}(Y \mid X)] = \mathbb{E}Y.$$

Since a function of X is known when conditioning on X ,

$$\mathbb{E}[g(X)Y \mid X] = g(X)\mathbb{E}(Y \mid X).$$

Consequently,

$$\mathbb{E}u = \mathbb{E}[\mathbb{E}(u \mid X)] = 0,$$

and

$$\mathbb{E}[g(X)u] = \mathbb{E}\{g(X)\mathbb{E}(u \mid X)\} = 0,$$

provided the relevant expectations exist.

Thus conditional mean independence is stronger than uncorrelatedness with only the linear regressors.

2 Introduction to Econometrics and Causality

Source: Section 1.1

Question 9: What econometrics tries to accomplish

Identify the main purposes of econometrics. Using the school-class-size example, distinguish description, prediction, and causal interpretation.

Worked solution

Econometrics is used to test economic theories, evaluate policies, quantify relationships, and make predictions.

Finding that districts with smaller classes have higher average test scores is descriptive.

Using class size and other variables to predict a district's future score is predictive.

Claiming that hiring teachers to reduce class size would raise test scores is causal.

The last claim requires more than an observed correlation: income, student composition, resources, and other factors may differ across districts.

Sampling uncertainty and causal identification are separate problems. A precisely estimated association can still be causally misleading.

Question 10: Randomized experiments and observational data

Explain how random assignment helps identify a treatment effect. Why can spillovers undermine the fertilizer experiment in the notes? Why use observational data?

Worked solution

Random assignment makes treatment independent of potential outcomes in the assignment design. Treatment and control groups are therefore comparable in expectation, although finite-sample imbalances remain possible.

In the fertilizer example, nutrients may reach neighboring untreated plots. Then a plot's outcome depends on other plots' treatment, violating the simple no-interference setup.

Observational data are often used because experiments are expensive, impractical, legally constrained, or ethically unacceptable. They can cover large populations and naturally occurring behavior.

Their disadvantage is that treatment or exposure may be related to other outcome determinants. A research design must address this selection rather than merely calculate an association.

Question 11: Cross sections, time series, and panels

Define the three main data structures in the notes. When does a panel have NT observations?

Worked solution

A cross section observes different units, usually at one period:

$$(y_i, x_i), \quad i = 1, \dots, N.$$

A time series follows a unit or aggregate over time:

$$(y_t, x_t), \quad t = 1, \dots, T.$$

Dependence across dates is common, so independent sampling cannot simply be assumed.

A panel follows multiple units over time:

$$(y_{it}, x_{it}).$$

A balanced panel with every unit observed in every period has NT observations. An unbalanced panel has missing unit-period combinations.

Repeated cross sections need not follow the same individuals, so they are not automatically a panel.

3 Simple Linear Regression

Source: Section 1.2

Question 12: Population regression and sample regression

Distinguish the conditional expectation, population error, fitted value, and residual in a simple regression.

Worked solution

The population model is

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

Under $\mathbb{E}(u_i | x_i) = 0$,

$$\mathbb{E}(y_i | x_i) = \beta_0 + \beta_1 x_i.$$

The fitted value is

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i,$$

and the residual is

$$\hat{u}_i = y_i - \hat{y}_i.$$

The population error uses unknown population coefficients; the residual uses estimated coefficients.

For

$$\mathbb{E}(\text{score} | \text{class size}) = 720 - 0.6 \text{ class size},$$

a class size of 20 gives expected score 708. An actual score can differ from 708.

The intercept may refer to an unrealistic regressor value; its algebraic role does not guarantee a meaningful extrapolation to $x = 0$.

Question 13: Deriving the simple OLS coefficients

Derive the OLS intercept and slope from the first-order conditions.

Worked solution

Minimize

$$S(b_0, b_1) = \sum_i (y_i - b_0 - b_1 x_i)^2.$$

The first-order conditions are

$$\sum_i (y_i - b_0 - b_1 x_i) = 0,$$

$$\sum_i x_i (y_i - b_0 - b_1 x_i) = 0.$$

The first gives

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Substituting into the second,

$$\sum_i x_i (y_i - \bar{y}) = \hat{\beta}_1 \sum_i x_i (x_i - \bar{x}).$$

Using the summation identities,

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}.$$

Therefore,

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The denominator must be positive. If every x_i is identical, the intercept and slope cannot be separately identified.

Question 14: The method-of-moments interpretation

Show how the moment conditions generate the same estimator as least squares. Do sample normal equations prove population exogeneity?

Worked solution

Suppose

$$\mathbb{E}u_i = 0, \quad \mathbb{E}(x_i u_i) = 0.$$

Replace these expectations with sample averages:

$$\frac{1}{N} \sum_i (y_i - b_0 - b_1 x_i) = 0,$$

$$\frac{1}{N} \sum_i x_i (y_i - b_0 - b_1 x_i) = 0.$$

These are the OLS normal equations, so solving them gives the same coefficients.

However, OLS enforces these sample relationships by construction. Observing

$$\sum_i x_i \hat{u}_i = 0$$

does not prove that the unobserved population errors satisfy $\mathbb{E}(x_i u_i) = 0$.

The algebra of the estimator and the assumptions needed for its interpretation are different matters.

Question 15: Algebraic properties of OLS

For an OLS regression with an intercept, prove that residuals sum to zero, fitted values have the same mean as outcomes, and the regression line passes through the sample means.

Worked solution

The intercept normal equation gives

$$\sum_i \hat{u}_i = 0.$$

Since $y_i = \hat{y}_i + \hat{u}_i$,

$$\bar{y} = \bar{\hat{y}}.$$

Also,

$$\bar{\hat{y}} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x},$$

so the fitted line passes through $(\bar{x}, \bar{y})$.

The slope normal equation gives

$$\sum_i x_i \hat{u}_i = 0.$$

Together with the zero residual mean,

$$\sum_i (x_i - \bar{x}) \hat{u}_i = 0.$$

These are finite-sample algebraic properties and do not require that the regression be causally valid.

Question 16: Variation decomposition and goodness of fit

Derive $SST = SSE + SSR$ and explain the limitations of R^2 . Why is plotting the data still useful?

Worked solution

With an intercept,

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{u}_i.$$

Squaring and summing,

$$SST = SSE + SSR + 2 \sum_i (\hat{y}_i - \bar{y}) \hat{u}_i.$$

The cross term is zero because fitted values are linear combinations of regressors and residuals are orthogonal to all included regressors.

Thus

$$SST = SSE + SSR, \quad R^2 = 1 - \frac{SSR}{SST},$$

assuming $SST > 0$.

A high R^2 does not establish causality or correct functional form. A low R^2 does not rule out precise estimation of an important effect.

Anscombe's quartet in the notes illustrates that similar regression summaries can conceal very different patterns, outliers, and nonlinearities. Graphical inspection remains valuable.

Question 17: A complete numerical OLS example

For

$$x = (1, 2, 3), \quad y = (1, 2, 2),$$

calculate the OLS coefficients, fitted values, residuals, SSR , and R^2 .

Worked solution

The sample means are

$$\bar{x} = 2, \quad \bar{y} = \frac{5}{3}.$$

The centered sums are

$$\begin{aligned} S_{xx} &= (-1)^2 + 0^2 + 1^2 = 2, \\ S_{xy} &= (-1) \left(-\frac{2}{3}\right) + 0 \left(\frac{1}{3}\right) + 1 \left(\frac{1}{3}\right) = 1. \end{aligned}$$

Hence

$$\hat{\beta}_1 = \frac{1}{2}, \quad \hat{\beta}_0 = \frac{5}{3} - \frac{1}{2}(2) = \frac{2}{3}.$$

Fitted values are

$$\hat{y} = \left(\frac{7}{6}, \frac{5}{3}, \frac{13}{6}\right).$$

Residuals are

$$\hat{u} = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right).$$

They sum to zero.

Finally,

$$SSR = \frac{1}{36} + \frac{1}{9} + \frac{1}{36} = \frac{1}{6},$$

$$SST = \frac{4}{9} + \frac{1}{9} + \frac{1}{9} = \frac{2}{3}.$$

Therefore,

$$R^2 = 1 - \frac{1/6}{2/3} = \frac{3}{4}.$$

Question 18: Four functional forms

Interpret the slope in level–level, level–log, log–level, and log–log regressions.

Worked solution

For the systematic fitted relationship:

(a) Level–level:

$$y = \beta_0 + \beta_1 x + u.$$

A change Δx is associated with $\Delta \hat{y} = \beta_1 \Delta x$.

(b) Level–log:

$$y = \beta_0 + \beta_1 \log x + u.$$

A 1% increase in x changes fitted y by approximately $\beta_1/100$ units. For a proportional change r , the exact change is $\beta_1 \log(1 + r)$.

(c) Log–level:

$$\log y = \beta_0 + \beta_1 x + u.$$

A unit increase in x changes fitted log outcome by β_1 , approximately $100\beta_1\%$ in the exponentiated fitted outcome. The exact percentage is $100(e^{\beta_1} - 1)$.

(d) Log–log:

$$\log y = \beta_0 + \beta_1 \log x + u.$$

β_1 is an elasticity. A 1% increase in x corresponds approximately to a $\beta_1\%$ increase in the exponentiated fitted outcome.

For logged outcomes, interpreting these statements as changes in the arithmetic conditional mean of y needs additional retransformation assumptions.

Question 19: Simple-regression assumptions

State SLR.1–SLR.5 and identify which are needed for unbiasedness and which additional assumption gives BLUE.

Worked solution

The assumptions are:

- (a) Linearity in parameters: $y_i = \beta_0 + \beta_1 x_i + u_i$.
- (b) Random sampling of observation pairs.
- (c) Sample variation in x .
- (d) Zero conditional mean: $\mathbb{E}(u_i | x_i) = 0$.
- (e) Homoskedasticity: $\text{Var}(u_i | x_i) = \sigma^2$.

Under the usual random-sampling formulation, the first four give conditional and unconditional unbiasedness.

Adding homoskedasticity yields the Gauss–Markov result: OLS is the best linear unbiased estimator.

Normal errors are not required for BLUE. They are an additional assumption used for exact small-sample distributional inference.

A transformed regressor such as $\log x$ is compatible with linearity in parameters.

Question 20: Proving unbiasedness

Prove that both simple-regression OLS coefficients are unbiased under the appropriate assumptions.

Worked solution

Let

$$S_{xx} = \sum_i (x_i - \bar{x})^2.$$

Substituting the population model into the slope formula,

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_i (x_i - \bar{x}) u_i}{S_{xx}}.$$

Condition on the full observed regressor sample $\mathbf{x}$. Under random sampling and zero conditional mean,

$$\mathbb{E}(u_i | \mathbf{x}) = 0.$$

Therefore,

$$\mathbb{E}(\hat{\beta}_1 | \mathbf{x}) = \beta_1.$$

For the intercept,

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Since

$$\mathbb{E}(\bar{y} | \mathbf{x}) = \beta_0 + \beta_1 \bar{x},$$

we get

$$\mathbb{E}(\hat{\beta}_0 | \mathbf{x}) = \beta_0.$$

Applying iterated expectations gives unconditional unbiasedness as well.

Question 21: Coefficient variances and standard errors

Derive the slope variance, give the intercept variance and covariance, and explain the estimator of σ^2 .

Worked solution

Define

$$w_i = \frac{x_i - \bar{x}}{S_{xx}}, \quad \hat{\beta}_1 - \beta_1 = \sum_i w_i u_i.$$

Under homoskedastic uncorrelated errors,

$$\text{Var}(\hat{\beta}_1 \mid \mathbf{x}) = \sigma^2 \sum_i w_i^2 = \boxed{\frac{\sigma^2}{S_{xx}}}.$$

The other formulas are

$$\text{Var}(\hat{\beta}_0 \mid \mathbf{x}) = \sigma^2 \left(\frac{1}{N} + \frac{\bar{x}^2}{S_{xx}} \right),$$

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1 \mid \mathbf{x}) = -\frac{\sigma^2 \bar{x}}{S_{xx}}.$$

An unbiased variance estimator is

$$s^2 = \frac{SSR}{N-2}.$$

Two degrees of freedom are used to estimate the intercept and slope.

The standard error of regression is $s = \sqrt{s^2}$. The slope standard error is

$$\text{se}(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}.$$

The square root of an unbiased variance estimator is not generally unbiased for the standard deviation.

Question 22: Proving the Gauss–Markov efficiency result

Show why no other linear unbiased slope estimator has smaller conditional variance than OLS.

Worked solution

Consider another linear estimator

$$\tilde{\beta}_1 = \sum_i a_i y_i.$$

For unbiasedness at every β_0, β_1 , its weights must satisfy

$$\sum_i a_i = 0, \quad \sum_i a_i x_i = 1.$$

Write $a_i = w_i + d_i$, where w_i are the OLS weights. Then

$$\sum_i d_i = 0, \quad \sum_i d_i x_i = 0.$$

Consequently,

$$\sum_i w_i d_i = \frac{\sum_i (x_i - \bar{x}) d_i}{S_{xx}} = 0.$$

Under homoskedasticity,

$$\text{Var}(\tilde{\beta}_1 \mid \mathbf{x}) = \sigma^2 \sum_i (w_i + d_i)^2$$

$$\begin{aligned}
&= \sigma^2 \sum_i w_i^2 + \sigma^2 \sum_i d_i^2 \\
&\geq \text{Var}(\hat{\beta}_1 \mid \mathbf{x}).
\end{aligned}$$

The comparison is within the class of linear unbiased estimators. It is not a universal claim about all estimators under every loss function.

Question 23: Regression with only one parameter

Derive the no-intercept slope and the intercept-only estimator. What happens if an intercept is incorrectly omitted?

Worked solution

For $y_i = b_1 x_i + u_i$, minimizing squared residuals gives

$$\hat{b}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2}.$$

If the true model includes an intercept,

$$y_i = \beta_0 + \beta_1 x_i + u_i,$$

then

$$\hat{b}_1 = \beta_1 + \beta_0 \frac{\sum_i x_i}{\sum_i x_i^2} + \frac{\sum_i x_i u_i}{\sum_i x_i^2}.$$

Under zero conditional mean,

$$\mathbb{E}(\hat{b}_1 \mid \mathbf{x}) - \beta_1 = \beta_0 \frac{\sum_i x_i}{\sum_i x_i^2}.$$

The bias vanishes if $\beta_0 = 0$ or the sample x values sum to zero.

For the intercept-only model,

$$\min_b \sum_i (y_i - b)^2$$

gives

$$\hat{b} = \bar{y}.$$

Without an intercept, residuals need not sum to zero and the usual centered R^2 decomposition need not hold.

Question 24: Binary regressors and treatment effects

Interpret regression on a treatment dummy. Define potential outcomes, ATE, and the fundamental problem of causal inference.

Worked solution

For

$$y_i = \beta_0 + \beta_1 D_i + u_i, \quad D_i \in \{0, 1\},$$

OLS gives

$$\hat{\beta}_0 = \bar{y}_0, \quad \hat{\beta}_1 = \bar{y}_1 - \bar{y}_0,$$

provided both groups are observed.

Potential outcomes are $Y_i(1)$ and $Y_i(0)$. The observed outcome is

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

The individual effect is $Y_i(1) - Y_i(0)$, and

$$ATE = \mathbb{E}[Y_i(1) - Y_i(0)].$$

Only one potential outcome is observed per individual. The missing other outcome is the counterfactual: this is the fundamental problem of causal inference.

Under random assignment, consistency of observed outcomes, and the relevant no-interference conditions, the difference in group means identifies the ATE.

In observational data,

$$\begin{aligned} \mathbb{E}(Y \mid D = 1) - \mathbb{E}(Y \mid D = 0) &= \mathbb{E}[Y(1) - Y(0) \mid D = 1] \\ &\quad + \mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]. \end{aligned}$$

The second line is selection bias.

Defining a zero-mean regression residual within each group does not, by itself, establish a causal treatment effect.

4 Multiple Linear Regression

Source: Sections 1.3 and 1.5.1

Question 25: Multiple regression and ceteris paribus

Write the model in scalar, vector, and matrix form. Explain the meaning of a partial slope.

Worked solution

The scalar model is

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_K x_{iK} + u_i.$$

Let

$$\mathbf{x}_i = (1, x_{i1}, \dots, x_{iK})'.$$

Then

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + u_i.$$

Stacking observations gives

$$\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \mathbf{u},$$

where $\mathbf{X}$ is $N \times p$, $p = K + 1$.

Under a correctly specified linear conditional mean,

$$\frac{\partial \mathbb{E}(y_i \mid \mathbf{x}_i)}{\partial x_{ij}} = \beta_j,$$

holding the other underlying regressors fixed.

This is a conditional association. A causal interpretation requires suitable exogeneity and an appropriate control set.

With squares or interactions, a coefficient alone need not equal the partial effect of changing an underlying variable.

Question 26: Deriving matrix OLS

Derive matrix OLS and its normal equations. Why can the two cross terms in the squared-error objective be combined?

Worked solution

Expand

$$\begin{aligned} S(\mathbf{b}) &= (\mathbf{y} - X\mathbf{b})'(\mathbf{y} - X\mathbf{b}) \\ &= \mathbf{y}'\mathbf{y} - \mathbf{b}'X'\mathbf{y} - \mathbf{y}'X\mathbf{b} + \mathbf{b}'X'X\mathbf{b}. \end{aligned}$$

The cross terms are scalars and transposes of each other, so they are equal:

$$S(\mathbf{b}) = \mathbf{y}'\mathbf{y} - 2\mathbf{b}'X'\mathbf{y} + \mathbf{b}'X'X\mathbf{b}.$$

Differentiate:

$$\frac{\partial S}{\partial \mathbf{b}} = -2X'\mathbf{y} + 2X'X\mathbf{b}.$$

Thus

$$X'X\hat{\boldsymbol{\beta}} = X'\mathbf{y},$$

and, with full column rank,

$$\boxed{\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y}.}$$

Equivalently,

$$X'\hat{\mathbf{u}} = 0.$$

With an intercept, the fitted hyperplane passes through the sample means of the outcome and all regressors.

Question 27: Multiple-regression assumptions and unbiasedness

State the matrix conditions for unbiasedness and show the sampling-error decomposition. Distinguish perfect from high multicollinearity.

Worked solution

Assume a linear population model, random sampling, full column rank of X , and

$$\mathbb{E}(\mathbf{u} \mid X) = \mathbf{0}.$$

Then

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (X'X)^{-1}X'\mathbf{u},$$

so

$$\mathbb{E}(\hat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta}.$$

Perfect multicollinearity means an exact linear dependence among columns of X , preventing a unique OLS coefficient vector.

High but imperfect multicollinearity does not make OLS undefined or inherently biased. It can make individual coefficient estimates imprecise.

Including x and x^2 does not automatically violate rank. However, with a binary x , $x^2 = x$, so those two columns are exactly identical.

Question 28: Covariance matrices and variance inflation

Derive the homoskedastic OLS covariance matrix and interpret the individual-coefficient variance formula.

Worked solution

Under

$$\text{Var}(\mathbf{u} \mid X) = \sigma^2 I_N,$$

the sampling-error decomposition gives

$$\begin{aligned}\text{Var}(\hat{\beta} \mid X) &= (X'X)^{-1}X'(\sigma^2 I_N)X(X'X)^{-1} \\ &= \boxed{\sigma^2(X'X)^{-1}}.\end{aligned}$$

For a slope coefficient,

$$\text{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{\sum_i (x_{ij} - \bar{x}_j)^2 (1 - R_j^2)},$$

where R_j^2 comes from regressing x_j on the remaining regressors, including an intercept.

The variance inflation factor is

$$VIF_j = \frac{1}{1 - R_j^2}.$$

If $R_j^2 = 0.9$, the VIF is 10.

More unexplained noise increases variance. More independent variation in x_j improves precision.

Estimate the error variance using

$$s^2 = \frac{SSR}{N - p}.$$

The covariance matrix also contains off-diagonal elements, which are needed for tests of coefficient combinations.

Question 29: Frisch–Waugh–Lovell theorem

Explain partialling out and derive the FWL coefficient. What caution applies to standard errors from the final residual regression?

Worked solution

Partition the regression as

$$\mathbf{y} = \mathbf{x}\beta_1 + Z\gamma + \mathbf{u},$$

where Z includes the controls and intercept.

Define

$$M_Z = I - Z(Z'Z)^{-1}Z'.$$

Residualize both variables:

$$\mathbf{r}_y = M_Z \mathbf{y}, \quad \mathbf{r}_x = M_Z \mathbf{x}.$$

OLS in the full model gives

$$\boxed{\hat{\beta}_1 = (\mathbf{x}'M_Z\mathbf{x})^{-1}\mathbf{x}'M_Z\mathbf{y} = \frac{\mathbf{r}_x'\mathbf{r}_y}{\mathbf{r}_x'\mathbf{r}_x}}.$$

Thus the slope relates the part of x unexplained by controls to the part of y unexplained by controls.

The coefficient and residuals match the full regression. However, a naive final residual regression can use an incorrect residual degrees-of-freedom count. Conventional standard errors must account for the controls removed in the earlier steps.

FWL is an algebraic result; partialling out controls does not automatically establish causality.

Question 30: Omitted-variable bias

Derive omitted-variable bias in the two-regressor model and in matrix form. Give a sign example.

Worked solution

Suppose

$$y = \beta_0 + \beta_1 x + \beta_2 z + u,$$

and the population linear projection of z on x is

$$z = \delta_0 + \delta_1 x + v.$$

Substitution gives

$$y = (\beta_0 + \beta_2 \delta_0) + (\beta_1 + \beta_2 \delta_1)x + (u + \beta_2 v).$$

Thus the short-regression slope targets

$$\boxed{\beta_1^{\text{short}} = \beta_1 + \beta_2 \delta_1}.$$

Since

$$\delta_1 = \frac{\text{Cov}(x, z)}{\text{Var}(x)},$$

the omitted-variable bias is

$$\beta_2 \frac{\text{Cov}(x, z)}{\text{Var}(x)}.$$

If ability raises wages and is positively related to education, omitting ability tends to bias the education slope upward under this model.

In matrix form, if

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \mathbf{u}$$

but only $\mathbf{X}$ is included, then

$$\hat{\boldsymbol{\beta}}_S = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\boldsymbol{\gamma} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}.$$

Conditional on $\mathbf{X}, \mathbf{Z}$, the middle term is the bias under zero conditional mean of $\mathbf{u}$.

Question 31: Irrelevant variables and bad controls

Does adding an irrelevant variable necessarily create bias? Why can adding a control nevertheless be harmful?

Worked solution

No. If the added variable has true coefficient zero and the enlarged regressor set still satisfies zero conditional mean, OLS remains unbiased.

Costs can include higher variance, fewer residual degrees of freedom, and unstable estimates from multicollinearity.

A bad control is different. For example, controlling for a post-treatment mediator can remove part of the total treatment effect. Conditioning on a collider can create a spurious association.

Thus adding variables can change the estimand or violate identification assumptions. But ordinary overspecification with exogenous irrelevant variables does not itself imply bias.

An increase in R^2 does not distinguish useful controls from harmful ones.

Question 32: Interpreting the earnings example

The notes report approximately

$$\log(\widehat{\text{earnings}}) = 1.2207 + 0.09413 \text{ education} - 0.23387 \text{ female} + 0.008869 \text{ age}.$$

Interpret the slopes and the intercept carefully.

Worked solution

Holding age and the gender indicator fixed, one additional year of education is associated with 0.09413 higher fitted log earnings.

The approximate percentage difference is 9.413%; the exact difference in exponentiated fitted earnings is

$$100(e^{0.09413} - 1) \approx 9.87\%.$$

The female coefficient corresponds to

$$100(e^{-0.23387} - 1) \approx -20.85\%$$

relative to the reference category, holding the other included variables fixed. The approximation -23.39% is noticeably different.

The age coefficient corresponds approximately to 0.887% per additional year.

The intercept describes fitted log earnings at zero education, zero age, and the reference gender category. This is largely an extrapolation.

Exponentiating the fitted log value does not automatically give the arithmetic conditional mean of earnings.

Question 33: Consistency and weaker exogeneity

Prove consistency using normalized sample moments. How does consistency differ from unbiasedness?

Worked solution

Write

$$\hat{\beta} - \beta = \left(\frac{X'X}{N} \right)^{-1} \left(\frac{X'u}{N} \right).$$

Assume suitable laws of large numbers, finite moments,

$$\frac{X'X}{N} \xrightarrow{p} Q$$

with nonsingular Q , and

$$\frac{X'u}{N} \xrightarrow{p} \mathbb{E}(x_i u_i) = \mathbf{0}.$$

Then

$$\boxed{\hat{\beta} \xrightarrow{p} \beta}.$$

Consistency means convergence in probability as sample size grows. Unbiasedness means that the finite-sample expectation equals the parameter.

The weaker moment restriction $\mathbb{E}(\mathbf{x}_i u_i) = 0$ can establish consistency without the stronger condition $\mathbb{E}(u_i | \mathbf{x}_i) = 0$. Finite-sample unbiasedness is then not guaranteed.

Question 34: Endogeneity and inconsistency

Give the probability limit of OLS when exogeneity fails. Does a larger sample remove the resulting problem?

Worked solution

If

$$\frac{X'X}{N} \xrightarrow{p} Q, \quad \frac{X'u}{N} \xrightarrow{p} g,$$

then

$$\boxed{\text{plim } \hat{\beta} = \beta + Q^{-1}g}.$$

If $g \neq 0$, OLS is generally inconsistent.

For simple regression with an intercept,

$$\text{plim } \hat{\beta}_1 = \beta_1 + \frac{\text{Cov}(x, u)}{\text{Var}(x)}.$$

With multiple controls, the corresponding scalar expression uses the component of x_j residualized on the other regressors:

$$\text{plim } \hat{\beta}_j = \beta_j + \frac{\mathbb{E}(r_j u)}{\mathbb{E}(r_j^2)}.$$

The raw covariance ratio generally does not apply unchanged to a multiple-regression coefficient. A larger sample reduces sampling noise but does not eliminate a nonzero probability-limit bias.

5 Testing and Inference

Source: Section 1.4

Question 35: Normality and exact small-sample inference

What extra assumption gives an exact normal distribution for OLS? Explain why estimating σ produces a t distribution.

Worked solution

Assume

$$\mathbf{u} | X \sim N(\mathbf{0}, \sigma^2 I_N).$$

Then

$$\hat{\beta} | X \sim N\left(\beta, \sigma^2 (X'X)^{-1}\right).$$

In the normal linear model, OLS is also minimum-variance unbiased in the broader unbiased-estimator class, subject to the usual fixed-design conditions.

If σ were known, a standardized coefficient would be standard normal.

With

$$s^2 = \frac{SSR}{N-p},$$

we have

$$\frac{(N-p)s^2}{\sigma^2} \sim \chi_{N-p}^2,$$

independently of the OLS coefficient estimates.

Combining the normal numerator and independent estimated scale gives

$$\frac{\hat{\beta}_j - \beta_j}{\text{se}(\hat{\beta}_j)} \sim t_{N-p}.$$

Homoskedasticity alone does not establish an exact small-sample t distribution.

Question 36: A two-sided coefficient test

Suppose $\hat{\beta}_1 = 0.12$, $\text{se}(\hat{\beta}_1) = 0.04$, and residual degrees of freedom are 25. Test $H_0 : \beta_1 = 0$ at 5%, using critical value 2.060.

Worked solution

The alternative is $H_1 : \beta_1 \neq 0$. The statistic is

$$t = \frac{0.12 - 0}{0.04} = 3.$$

Since

$$|3| > 2.060,$$

reject the null at 5%.

The estimate is statistically distinguishable from zero under the assumed model and inference procedure.

This does not prove that the association is causal, nor does it establish that the magnitude is economically important.

Question 37: One-sided and nonzero-null tests

Using the same estimate and standard error, test $H_0 : \beta_1 \leq 0.05$ against $H_1 : \beta_1 > 0.05$. Use the one-sided 5% critical value 1.708 for 25 degrees of freedom.

Worked solution

Evaluate the statistic at the null boundary:

$$t = \frac{0.12 - 0.05}{0.04} = 1.75.$$

Since $1.75 > 1.708$, reject the one-sided null at 5%.

For the two-sided test

$$H_0 : \beta_1 = 0.05, \quad H_1 : \beta_1 \neq 0.05,$$

the same statistic does not exceed 2.060, so we fail to reject.

The direction of a one-sided test should be chosen from the research question before inspecting the estimate.

Question 38: P-values, error types, and practical importance

Define a p -value, significance level, Type I error, Type II error, and power. What does $p = 0.03$ not mean?

Worked solution

A p -value is the probability, under the null model, of a test statistic at least as extreme as the observed one in the direction specified by the alternative.

For a two-sided t test,

$$p = 2P(t_\nu \geq |t_{\text{obs}}|).$$

The significance level α controls the probability of rejecting a true null for a valid test.

A Type I error rejects a true null. A Type II error fails to reject a false null. Power is the probability of rejecting under a specified alternative.

A p -value of 0.03 is not a 3% probability that the null is true. It is not the probability that the result occurred “by chance.”

Statistical significance concerns evidence relative to uncertainty. Practical significance concerns magnitude, units, costs, and the substantive question.

Question 39: Confidence intervals

Construct a 95% confidence interval from Question 36. Explain its repeated-sampling meaning.

Worked solution

The interval is

$$\hat{\beta}_1 \pm t_{25,0.975} \text{se}(\hat{\beta}_1) = 0.12 \pm 2.060(0.04).$$

Thus

$$\boxed{[0.0376, 0.2024]}.$$

If the sampling procedure were repeated and an interval constructed each time, 95% of those intervals would cover the fixed parameter under the assumed model.

In frequentist inference, the parameter is fixed and the interval is random before observing the sample.

A two-sided 5% test rejects a particular null value exactly when that value lies outside the corresponding 95% interval.

Question 40: Testing a coefficient difference

How would you test $H_0 : \beta_1 = \beta_2$? Why are the two separate standard errors insufficient?

Worked solution

Rewrite the restriction as

$$H_0 : \beta_1 - \beta_2 = 0.$$

The estimated difference is

$$\hat{d} = \hat{\beta}_1 - \hat{\beta}_2.$$

Its variance is

$$\widehat{\text{Var}}(\hat{d}) = \widehat{\text{Var}}(\hat{\beta}_1) + \widehat{\text{Var}}(\hat{\beta}_2) - 2\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2).$$

The test statistic is

$$t = \frac{\hat{d}}{\sqrt{\widehat{\text{Var}}(\hat{d})}}.$$

For example, if the two variances are 0.04 and 0.09, and their covariance is 0.03, then

$$\text{se}(\hat{d}) = \sqrt{0.04 + 0.09 - 0.06} = \sqrt{0.07}.$$

Ignoring covariance generally produces an incorrect test. Use the covariance estimator appropriate to the error assumptions.

Question 41: Joint restrictions and the F test

State the restricted-versus-unrestricted F statistic, the global F statistic, and the relationship between a single-restriction F test and a t test.

Worked solution

For q restrictions on an unrestricted model with p parameters,

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(N - p)}.$$

Under the normal homoskedastic linear model,

$$F \sim F_{q, N-p}$$

under the null.

For compatible regressions using the same outcome and sample,

$$F = \frac{(R_U^2 - R_R^2)/q}{(1 - R_U^2)/(N - p)}.$$

For testing all K slopes jointly against zero, the restricted model contains only an intercept:

$$F_{\text{global}} = \frac{R^2/K}{(1 - R^2)/(N - K - 1)}.$$

For one restriction,

$$F = t^2,$$

so the F test agrees with the corresponding two-sided t test.

The usual SSR-based formula is not generally a heteroskedasticity-robust test.

Question 42: The baseball joint-significance example

In the notes, the restricted and unrestricted baseball models have

$$SSR_R = 198.31, \quad SSR_U = 183.19,$$

with $q = 3$ restrictions and 347 unrestricted residual degrees of freedom. Calculate and interpret the F statistic.

Worked solution

The statistic is

$$F = \frac{(198.31 - 183.19)/3}{183.19/347} \approx 9.55.$$

The notes report a very small p -value, approximately 4.47×10^{-6} , so the three restrictions are jointly rejected.

The restricted variables were batting average, home runs per year, and runs batted in per year. Although their individual t tests were insignificant at conventional levels, they were jointly significant.

Correlated regressors can make separate coefficients imprecise while the group still contributes substantial explanatory information.

Question 43: The LM test

Describe the auxiliary-regression LM test for exclusion restrictions. If $N = 200$ and the auxiliary $R^2 = 0.04$, calculate the statistic.

Worked solution

For the standard homoskedastic linear-model version:

- (a) Estimate the restricted model.
- (b) Save its residuals.
- (c) Regress those residuals on the full set of regressors.
- (d) Calculate

$$LM = NR_{\text{aux}}^2.$$

Under the null and regularity conditions,

$$LM \xrightarrow{d} \chi_q^2,$$

where q is the number of restrictions.

Here,

$$LM = 200(0.04) = 8.$$

For $q = 2$, the 5% critical value is about 5.991, so the null is rejected.

This is a large-sample test; its chi-square reference distribution is generally not an exact finite-sample result.

Question 44: Large-sample normality

Explain how inference can remain possible without normally distributed errors. State the asymptotic distribution of OLS.

Worked solution

Under suitable independent sampling, finite moments, exogeneity, and nonsingular

$$Q = \mathbb{E}(\mathbf{x}_i \mathbf{x}_i'),$$

a central limit theorem gives

$$\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(\mathbf{0}, Q^{-1}SQ^{-1}),$$

where

$$S = \mathbb{E}(u_i^2 \mathbf{x}_i \mathbf{x}_i').$$

Under homoskedasticity, $S = \sigma^2 Q$, giving

$$Q^{-1}SQ^{-1} = \sigma^2 Q^{-1}.$$

Appropriately studentized individual statistics are asymptotically standard normal. Joint Wald statistics are asymptotically chi-square.

Large samples do not make the original errors normal, nor do they repair endogeneity. They justify an approximation to the estimator's sampling distribution.

6 Scaling, Functional Forms, and Dummy Variables

Source: Section 1.5

Question 45: Changing measurement units

What happens to coefficients, standard errors, and tests when the outcome is multiplied by $c > 0$, or a regressor is multiplied by $a > 0$?

Worked solution

If $y^* = cy$, then

$$y^* = c\beta_0 + \sum_j c\beta_j x_j + cu.$$

All coefficient estimates and their standard errors are multiplied by c .

If $x_j^* = ax_j$, then

$$\beta_j x_j = \frac{\beta_j}{a} x_j^*.$$

Its coefficient and standard error are divided by a .

For equivalent null hypotheses, standardized tests and p -values are unchanged. R^2 is also unchanged.

In particular, dividing the outcome by 10 divides all coefficients and residuals by 10. It does not multiply them by 10.

With squares and interactions, every affected transformed column must be rescaled consistently.

Question 46: Exact log changes and retransformation

For

$$\log y = \beta_0 + \beta_1 x + u,$$

derive the exact percentage change in exponentiated fitted values. Why is exponentiating fitted log outcome not generally the conditional mean of y ?

Worked solution

An increase Δx changes the systematic log component by $\beta_1 \Delta x$. The ratio of exponentiated fitted values is

$$e^{\beta_1 \Delta x}.$$

The percentage change is therefore

$$\boxed{100(e^{\beta_1 \Delta x} - 1)}.$$

But

$$\mathbb{E}(y \mid x) = e^{\beta_0 + \beta_1 x} \mathbb{E}(e^u \mid x).$$

The extra factor generally differs from one and may depend on x .

If $u \mid x \sim N(0, \sigma^2)$ with constant variance,

$$\mathbb{E}(y \mid x) = e^{\beta_0 + \beta_1 x + \sigma^2/2}.$$

If the retransformation factor is constant in x , it cancels in ratios of conditional means.

Logs can reduce skewness or moderate scale differences, but they do not guarantee homoskedasticity or a correct functional form. Values close to zero can also become extreme negative log values.

Question 47: Quadratic effects and turning points

For

$$m(x) = \beta_0 + \beta_1 x + \beta_2 x^2,$$

derive the marginal effect, the exact finite change, and the turning point. Apply them to $m(x) = 5 + 4x - 0.2x^2$.

Worked solution

The derivative is

$$m'(x) = \beta_1 + 2\beta_2 x.$$

For a finite increase h ,

$$m(x+h) - m(x) = (\beta_1 + 2\beta_2 x)h + \beta_2 h^2.$$

The derivative-times-change approximation omits $\beta_2 h^2$.

If $\beta_2 \neq 0$, the turning point is

$$x^* = -\frac{\beta_1}{2\beta_2}.$$

It is a maximum when $\beta_2 < 0$ and a minimum when $\beta_2 > 0$.

For the numerical example,

$$m'(x) = 4 - 0.4x, \quad x^* = 10.$$

At $x = 5$, the marginal effect is 2. The exact change from 5 to 6 is

$$2(1) - 0.2(1)^2 = 1.8.$$

A turning point outside the observed range should not be treated as well-supported evidence about the data.

Question 48: Interactions and group-specific slopes

Interpret

$$m(x, z) = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 xz.$$

Then interpret the case where $z = D$ is binary.

Worked solution

For continuous variables,

$$\frac{\partial m}{\partial x} = \beta_1 + \beta_3 z, \quad \frac{\partial m}{\partial z} = \beta_2 + \beta_3 x,$$

$$\frac{\partial^2 m}{\partial x \partial z} = \beta_3.$$

For $D = 0$,

$$m(x, 0) = \beta_0 + \beta_1 x.$$

For $D = 1$,

$$m(x, 1) = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)x.$$

Thus β_2 is the intercept difference and β_3 is the slope difference.

The group gap at a given x is

$$m(x, 1) - m(x, 0) = \beta_2 + \beta_3 x.$$

Testing equality of the two lines requires the joint null $\beta_2 = \beta_3 = 0$.

In the notes' wage example, the reference education slope is 0.53948, while the female slope is

$$0.53948 - 0.08600 = 0.45348.$$

These are conditional wage associations.

Question 49: Average partial effects

For

$$m(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_2^2 + \beta_4 x_1 x_2,$$

derive both average partial effects. Must an average partial effect differ from the partial effect evaluated at the means?

Worked solution

The individual partial effects are

$$PE_{1i} = \beta_1 + \beta_4 x_{i2},$$

$$PE_{2i} = \beta_2 + 2\beta_3 x_{i2} + \beta_4 x_{i1}.$$

Averaging fitted effects gives

$$\widehat{APE}_1 = \hat{\beta}_1 + \hat{\beta}_4 \bar{x}_2,$$

$$\widehat{APE}_2 = \hat{\beta}_2 + 2\hat{\beta}_3 \bar{x}_2 + \hat{\beta}_4 \bar{x}_1.$$

Here the derivatives are affine in the regressors, so the APE equals the derivative evaluated at sample means.

They need not agree when the derivative is nonlinear. For example, if $m(x) = \beta_0 + \beta_1 x^3$,

$$APE = 3\beta_1 \bar{x}^2, \quad PEA = 3\beta_1 \bar{x}^2.$$

In the attendance example, the attendance effect is

$$-0.006713 + 0.005586 \text{ prior GPA}.$$

Thus the coefficient on attendance alone is not its average partial effect.

Question 50: Adjusted R-squared

State adjusted R^2 and calculate it for $N = 100$, $K = 4$, and $R^2 = 0.40$. What model comparisons are inappropriate?

Worked solution

With an intercept,

$$\bar{R}^2 = 1 - \frac{SSR/(N - K - 1)}{SST/(N - 1)} = 1 - (1 - R^2) \frac{N - 1}{N - K - 1}.$$

Numerically,

$$\bar{R}^2 = 1 - 0.60 \frac{99}{95} \approx 0.3747.$$

Raw R^2 cannot decrease when regressors are added to a nested model using the same observations. Adjusted R^2 can decrease and can be negative.

Comparisons should use the same dependent variable and sample. Directly comparing the R^2 of a wage regression with that of a log-wage regression is not meaningful.

Adjusted R^2 is a descriptive model-comparison criterion, not proof of causal validity or a general hypothesis test.

Question 51: Categorical variables and the dummy trap

There are three exhaustive car-color categories: blue, red, and black. Explain two valid parameterizations and interpret a change of reference category.

Worked solution

Because

$$D_{\text{blue}} + D_{\text{red}} + D_{\text{black}} = 1,$$

including all three indicators plus an intercept creates perfect multicollinearity.

One valid parameterization is

$$y = \beta_0 + \beta_R D_{\text{red}} + \beta_B D_{\text{black}} + u.$$

Blue is the reference:

$$\mathbb{E}(y \mid \text{blue}) = \beta_0,$$

$$\mathbb{E}(y \mid \text{red}) = \beta_0 + \beta_R,$$

$$\mathbb{E}(y \mid \text{black}) = \beta_0 + \beta_B.$$

Another valid parameterization includes all three dummies but no separate intercept. Each coefficient is its group mean in this simple model.

Changing the reference category changes the coefficient interpretations but leaves fitted values and residuals unchanged.

With additional numerical regressors, these become group-specific intercepts evaluated where the numerical regressors equal zero.

Question 52: The linear probability model

Explain the LPM, its coefficient interpretation, its conditional variance, and its main limitation. Interpret the notes' coefficient -0.2618 on children under six.

Worked solution

If $Y \in \{0, 1\}$,

$$\mathbb{E}(Y | X) = P(Y = 1 | X).$$

The LPM specifies

$$P(Y = 1 | X) = X'\beta$$

on the relevant support.

A coefficient 0.04 means a four-percentage-point probability difference for a unit regressor change, holding the other included variables fixed.

Writing $p(X) = P(Y = 1 | X)$,

$$\text{Var}(u | X) = \text{Var}(Y | X) = p(X)[1 - p(X)].$$

This generally varies with X , so heteroskedasticity is inherent.

OLS predictions can fall outside $[0, 1]$, making them invalid probabilities.

In the standard `mroz` data, `kidslt6` is a count of children under six. A coefficient -0.2618 represents approximately 26.18 percentage points lower fitted participation probability per additional child, holding controls fixed. It is not a 26.18% proportional reduction.

7 Heteroskedasticity and Weighted Least Squares

Source: Section 1.6

Question 53: Consequences of heteroskedasticity

Define heteroskedasticity. Which OLS properties survive, and which conventional claims fail?

Worked solution

Heteroskedasticity means

$$\text{Var}(u_i | X) = \sigma_i^2$$

varies across observations or regressor values.

With zero conditional mean and full rank, OLS remains conditionally unbiased. With appropriate moment and sampling conditions, it remains consistent.

But

$$\sigma^2(X'X)^{-1}$$

is generally no longer the correct covariance formula. Conventional standard errors and tests can therefore be invalid.

OLS is no longer guaranteed to be efficient among linear unbiased estimators.

A large sample alone does not make the conventional homoskedastic covariance estimator valid.

Question 54: The sandwich covariance formula

Derive the conditional covariance of OLS when $\text{Var}(\mathbf{u} \mid X) = \Omega$. Explain the HC0 estimator.

Worked solution

From

$$\hat{\beta} - \beta = (X'X)^{-1}X'\mathbf{u},$$

we obtain

$$\text{Var}(\hat{\beta} \mid X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

For uncorrelated heteroskedastic errors,

$$\Omega = \text{diag}(\sigma_1^2, \dots, \sigma_N^2).$$

HC0 replaces the covariance contribution with squared residuals:

$$\hat{V}_{\text{HC0}} = (X'X)^{-1} \left(\sum_i \mathbf{x}_i \mathbf{x}_i' \hat{u}_i^2 \right) (X'X)^{-1}.$$

Under regularity conditions, the resulting sandwich estimator consistently estimates the relevant asymptotic covariance.

This does not mean that each individual $\hat{u}_i^2$ consistently estimates its own σ_i^2 from one observation. Consistency concerns the aggregated covariance expression.

Question 55: HC1 and robust standard errors

What changes when conventional standard errors are replaced by HC1? Interpret the school example:

$$\hat{\beta} = -0.5482, \quad \text{se}_{\text{usual}} = 0.2296, \quad \text{se}_{\text{HC1}} = 0.2569.$$

Worked solution

With p fitted coefficients,

$$\hat{V}_{\text{HC1}} = \frac{N}{N-p} \hat{V}_{\text{HC0}}.$$

The OLS coefficients, fitted values, and residuals do not change. Their estimated uncertainty changes.

The two statistics are approximately

$$t_{\text{usual}} = -0.5482/0.2296 \approx -2.387,$$

$$t_{\text{HC1}} = -0.5482/0.2569 \approx -2.134.$$

The robust estimate of uncertainty is larger here, but robust standard errors need not always be larger.

Heteroskedasticity-robust inference is generally asymptotic. It does not automatically handle serial correlation or within-cluster dependence.

Question 56: Robust joint hypothesis testing

How can multiple restrictions be tested under heteroskedasticity?

Worked solution

Write the q independent linear restrictions as

$$H_0 : R\beta = r.$$

Using an appropriate robust covariance estimate $\widehat{V}$, form

$$W = (R\widehat{\beta} - r)'[R\widehat{V}R']^{-1}(R\widehat{\beta} - r).$$

Under the null and regularity conditions,

$$W \xrightarrow{d} \chi_q^2.$$

Some software reports W/q using an approximate F reference distribution.

The usual restricted-versus-unrestricted SSR F statistic does not automatically become robust simply because robust individual standard errors are also reported.

Question 57: Breusch–Pagan testing

Describe the residual auxiliary-regression procedure. If $N = 150$, the auxiliary regression has three nonconstant regressors, and $R_{\text{aux}}^2 = 0.06$, calculate the test statistic.

Worked solution

First estimate the main model and save residuals. Then regress

$$\widehat{u}_i^2 = \delta_0 + Z_i'\delta + v_i.$$

The null is

$$H_0 : \delta = \mathbf{0}.$$

The auxiliary-regression, studentized BP/Koenker-style statistic used in the notes is

$$LM = NR_{\text{aux}}^2.$$

Under suitable null conditions,

$$LM \xrightarrow{d} \chi_q^2,$$

where q is the number of nonconstant independent variance regressors.

Here,

$$LM = 150(0.06) = 9.$$

The 5% χ_3^2 critical value is approximately 7.815, so reject the null.

The test examines variance patterns related to the chosen auxiliary regressors. Failure to reject does not prove homoskedasticity.

Question 58: White tests and degrees of freedom

For two continuous regressors, write the full White auxiliary regression. State the degrees of freedom. How does the fitted-value alternative differ?

Worked solution

The full auxiliary regression is

$$\hat{u}_i^2 = \delta_0 + \delta_1 x_{i1} + \delta_2 x_{i2} + \delta_3 x_{i1}^2 + \delta_4 x_{i2}^2 + \delta_5 x_{i1} x_{i2} + v_i.$$

If these five nonconstant columns are independent,

$$NR_{\text{aux}}^2 \xrightarrow{d} \chi_5^2.$$

For K continuous regressors, the usual maximum number of nonconstant terms is

$$K + K + \frac{K(K-1)}{2} = \frac{K(K+3)}{2}.$$

Degrees of freedom equal the actual number of independent nonconstant auxiliary terms after removing duplicates or collinear columns.

A smaller alternative is

$$\hat{u}_i^2 = \delta_0 + \delta_1 \hat{y}_i + \delta_2 \hat{y}_i^2 + v_i.$$

It typically uses two degrees of freedom, but examines a more restricted set of variance patterns.

The full White test does not generally use only K degrees of freedom.

Question 59: Deriving weighted least squares

Suppose $\Omega = \text{diag}(\sigma_i^2)$ is known. Derive WLS by transforming the model. Distinguish the transformation divisor from the objective function weights.

Worked solution

Start with

$$y_i = \mathbf{x}_i' \beta + u_i.$$

Divide the entire equation, including its intercept column, by σ_i :

$$\frac{y_i}{\sigma_i} = \frac{\mathbf{x}_i'}{\sigma_i} \beta + \frac{u_i}{\sigma_i}.$$

Now

$$\text{Var}(u_i/\sigma_i \mid X) = 1.$$

In matrix notation,

$$\tilde{y} = \Omega^{-1/2} y, \quad \tilde{X} = \Omega^{-1/2} X.$$

OLS on the transformed model gives

$$\boxed{\hat{\beta}_{\text{WLS}} = (X' \Omega^{-1} X)^{-1} X' \Omega^{-1} y.}$$

Its conditional covariance is

$$\text{Var}(\hat{\beta}_{\text{WLS}} \mid X) = (X' \Omega^{-1} X)^{-1}.$$

Equivalently, WLS minimizes

$$\sum_i \frac{(y_i - \mathbf{x}_i' \beta)^2}{\sigma_i^2}.$$

Thus the equation is divided by σ_i , but the squared-error weight is $1/\sigma_i^2$.

GLS allows a general positive-definite Ω ; WLS is the diagonal-covariance case.

Question 60: Feasible GLS and its limitations

Describe feasible GLS based on a log-variance model. Is it guaranteed to outperform OLS if that variance model is wrong?

Worked solution

A common procedure is:

- (a) Fit the conditional-mean model by OLS.
- (b) Save residuals $\hat{u}_i$.
- (c) Model $\log(\hat{u}_i^2)$ using variance predictors.
- (d) Let $\hat{g}_i$ be the fitted log-scale values and set $\hat{h}_i = \exp(\hat{g}_i)$.
- (e) Refit the mean model using weights $w_i = 1/\hat{h}_i$.

The log transformation ensures positive fitted variance factors. Exactly zero residuals require special handling because their logarithms are undefined.

Under a suitable multiplicative error structure, the log-squared-error regression identifies relative variances up to a common scale; that scale does not affect WLS coefficients.

Correctly specified and consistently estimated weights can give asymptotic GLS efficiency.

If the variance model is wrong, FGLS is not guaranteed to be more efficient than OLS. With suitable exogeneity, positive well-behaved weights, and regularity conditions, the weighted estimator may remain consistent, but robust covariance estimation is advisable.

Estimated weights also mean exact finite-sample unbiasedness is not generally assured.

8 Practical R Exercises and Integrated Review

Question 61: Implementing regression and FWL in R

Write R code to construct the school variables, estimate a regression, and verify FWL on the same complete sample.

Worked solution

The following code uses the dataset employed in the notes.

```
library(AER)
data("CASchools")

d <- CASchools
d$str <- d$students / d$teachers
d$score <- (d$read + d$math) / 2

vars <- c("score", "str", "income", "english")
d <- d[complete.cases(d[, vars]), ]
```

```

full <- lm(score ~ str + income + english, data = d)
summary(full)

ry <- residuals(lm(score ~ income + english, data = d))
rx <- residuals(lm(str ~ income + english, data = d))

partial <- lm(ry ~ rx - 1)

c(full_slope = unname(coef(full)["str"]),
  FWL_slope = unname(coef(partial)["rx"]))

s2_full <- sum(residuals(full)^2) / df.residual(full)
FWL_se <- sqrt(s2_full / sum(rx^2))

c(full_se = summary(full)$coefficients["str", "Std. Error"],
  corrected_FWL_se = FWL_se)

```

Using the same complete sample matters: FWL is a same-sample algebraic identity.
The corrected FWL standard error uses the full model's residual degrees of freedom.

Question 62: Robust tests and feasible weighting in R

Write R code for HC1 inference, a robust joint test, heteroskedasticity tests, and a feasible weighted fit.

Worked solution

Continuing with the data constructed above:

```

library(sandwich)
library(lmtest)
library(car)

m <- lm(score ~ str + income + english, data = d)

V_HC1 <- vcovHC(m, type = "HC1")
coeftest(m, vcov. = V_HC1)

linearHypothesis(
  m,
  c("income = 0", "english = 0"),
  vcov. = V_HC1,
  test = "F"
)

# Studentized Breusch-Pagan test
bptest(m, studentize = TRUE)

# Fitted-value version of the White-style test
d$fit <- fitted(m)
bptest(
  m,

```

```

    ~ fit + I(fit^2),
    data = d,
    studentize = TRUE
)

# A feasible variance model
d$u2 <- residuals(m)^2

# Do not silently take log(0)
if (any(d$u2 <= 0)) {
  stop("Zero residual: revise the log-variance approach.")
}

variance_model <- lm(
  log(u2) ~ str + income + english,
  data = d
)

d$hhat <- exp(fitted(variance_model))

weighted <- lm(
  score ~ str + income + english,
  data = d,
  weights = 1 / hhat
)

coeftest(
  weighted,
  vcov. = vcovHC(weighted, type = "HC1")
)

```

In `lm`, the supplied weights multiply squared residuals. Therefore inverse variance, not inverse standard deviation, is used.

A fitted variance model is an assumption to assess, not a guaranteed improvement.

Question 63: Designing a defensible empirical analysis

You want to estimate the relationship between education and earnings. Describe a workflow covering identification, specification, estimation, inference, and reporting.

Worked solution

- Define the population and question. Decide whether the goal is prediction, conditional association, or a causal return to education.
- Inspect measurement units, missingness, sampling, and the support of earnings and education.
- Choose a functional form, such as log earnings, based on the economic relationship and data support.
- Explain the control set. Distinguish plausible confounders from mediators or other bad controls.

- (e) Estimate the model and inspect residual patterns, outliers, and multicollinearity.
- (f) Use a covariance estimator appropriate to the sampling and error dependence. Heteroskedasticity-robust standard errors are not a cure for endogeneity.
- (g) Report coefficient magnitudes, uncertainty, confidence intervals, sample size, and relevant joint tests.
- (h) Interpret log coefficients carefully and avoid confusing exponentiated fitted log earnings with mean earnings.
- (i) Compare defensible specifications without choosing solely by R^2 or significance stars.
- (j) State which assumptions support causal interpretation and which concerns remain unresolved.

Question 64: Identify and correct common mistakes

Explain why each statement is false or incomplete:

- (a) Zero correlation means independence.
- (b) OLS residual orthogonality proves exogeneity.
- (c) Adding an irrelevant variable always biases OLS.
- (d) Large samples solve heteroskedasticity automatically.
- (e) A full White test always has K degrees of freedom.
- (f) An LPM coefficient of -0.26 means a 26% reduction.
- (g) FGLS is always more efficient than OLS.
- (h) A small p -value proves a causal effect.

Worked solution

- (a) Zero correlation rules out linear association, but nonlinear dependence can remain.
- (b) Residual orthogonality is imposed by OLS algebra. Exogeneity concerns unobserved population errors.
- (c) Exogenous irrelevant regressors preserve unbiasedness, although they can reduce precision. Bad controls are a separate issue.
- (d) Heteroskedasticity can invalidate conventional standard errors even asymptotically. A suitable covariance estimator is needed.
- (e) The degrees of freedom equal the number of independent nonconstant terms in the auxiliary regression, including squares and interactions where applicable.
- (f) The interpretation is a 26-percentage-point change per unit regressor change. A proportional percentage change depends on the starting probability.
- (g) FGLS efficiency depends on the variance model and regularity conditions; misspecified weights need not help.
- (h) A small p -value concerns evidence against a statistical null under the assumed model. Causal identification requires additional reasoning.

Coverage Map

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Essential Formula Sheet

Simple OLS

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Multiple OLS

$$\hat{\beta} = (X'X)^{-1}X'y, \quad \hat{\beta} - \beta = (X'X)^{-1}X'u.$$

$$s^2 = \frac{SSR}{N - p}, \quad \hat{V}_{\text{usual}} = s^2(X'X)^{-1}.$$

Fit

$$SST = SSE + SSR, \quad R^2 = 1 - \frac{SSR}{SST}.$$

$$\bar{R}^2 = 1 - \frac{SSR/(N-p)}{SST/(N-1)}.$$

Inference

$$t = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{se}(\hat{\beta}_j)}.$$

$$CI_{1-\alpha} : \quad \hat{\beta}_j \pm t_{N-p, 1-\alpha/2} \text{se}(\hat{\beta}_j).$$

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(N-p)}.$$

$$LM = NR_{\text{aux}}^2.$$

Heteroskedasticity

$$\text{Var}(\hat{\beta} \mid X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

$$\hat{V}_{\text{HC0}} = (X'X)^{-1} \left(\sum_i x_i x_i' \hat{u}_i^2 \right) (X'X)^{-1}.$$

$$\hat{\beta}_{\text{WLS}} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y.$$

Interpretation

$$\text{Exact log-level percentage change} = 100(e^{\beta_j \Delta x_j} - 1).$$

$$\frac{\partial(\beta_0 + \beta_1 x + \beta_2 x^2)}{\partial x} = \beta_1 + 2\beta_2 x.$$

$$\frac{\partial(\beta_0 + \beta_1 x + \beta_2 z + \beta_3 xz)}{\partial x} = \beta_1 + \beta_3 z.$$